

Solutions – Practice Problem Set 6

ECON 101 – Macroeconomics

Fall 2025

Problem 1

The Bank of Canada observes that inflation is 3.8%, above the 2% target midpoint. It chooses **contractionary monetary policy**.

(a) Implementing contractionary policy with the key policy rate

Step 1: The policy instrument.

- The Bank of Canada's main policy instrument is the **target overnight interest rate** (the key policy rate).
- Contractionary (tight) monetary policy means the Bank *raises* this target rate.

Step 2: Operating band and settlement balances.

- The Bank sets an operating band around the target overnight rate (for example, ± 0.25 percentage points).
- The **Bank Rate** is at the top of the band and the **deposit rate** is at the bottom.
- To push the market overnight rate *up* toward its higher target, the Bank of Canada typically:
 - reduces the supply of settlement balances in the LVTS / payments system,
 - or drains liquidity so that banks are more likely to borrow at higher rates.

Step 3: Open-market operations.

- The Bank carries out **open-market sales** of government securities.
- Commercial banks and dealers pay for these securities, which reduces their cash/reserves.
- With fewer reserves, the supply of overnight funds falls, putting *upward* pressure on the overnight rate.
- The market overnight rate rises and aligns with the higher target rate.

Conclusion:

- Higher overnight interest rate $\Rightarrow$ tighter monetary conditions.
- This is the first step in contractionary monetary policy aimed at reducing inflation from 3.8% toward 2%.

(b) Monetary transmission mechanism (step-by-step)

Starting from a higher overnight interest rate:

Step 1: From overnight rate to other interest rates.

- Higher overnight rate increases short-term money market rates.
- Banks' funding costs rise, so they raise prime lending rates, mortgage rates, and other loan rates.
- Long-term bond yields may also rise as expectations adjust.

Step 2: Effect on consumption (C).

- Higher interest rates make borrowing for consumer durables (cars, appliances) and housing more expensive.
- Households reduce spending on credit-sensitive items.
- Wealth effects: higher rates may reduce asset prices (e.g., bonds, equities, housing), making households feel poorer and spend less.
- Overall, planned consumption spending falls.

Step 3: Effect on investment (I).

- Higher interest rates increase the cost of borrowing for firms.
- Some investment projects that were profitable at lower interest rates become unprofitable.
- Firms cut back on new capital spending (machinery, buildings, technology).
- Planned investment spending falls.

Step 4: Effect on exchange rate and net exports (NX).

- Higher Canadian interest rates attract foreign capital (higher return on CAD assets).
- Demand for Canadian dollars rises, causing the **Canadian dollar to appreciate**.
- An appreciation makes Canadian exports more expensive and imports cheaper.
- Net exports (NX) decline.

Step 5: Effect on aggregate demand (AD).

- Lower C , lower I , and lower NX all reduce total planned expenditure.
- In the AD–AS framework, this reduction appears as a **leftward shift of the AD curve**.

Step 6: Effect on real GDP and inflation.

- In the short run, with an upward-sloping SRAS, a leftward shift of AD reduces real GDP and the price level.
- Lower real GDP (below potential) increases unemployment.
- Excess capacity and weaker demand put downward pressure on inflation.
- Over time, inflation falls back toward the 2% target.

(c) AD–AS diagram and returning inflation to target

Step 1: Initial equilibrium.

- Initially, the economy is at a point where AD intersects SRAS and LRAS at an inflation rate of 3.8%.
- LRAS is vertical at potential GDP Y^* .

Step 2: Contractionary monetary policy.

- Higher policy interest rate shifts AD leftward (from AD_0 to AD_1).
- In the short run, the new equilibrium is at the intersection of AD_1 and SRAS.
- At this point:
 - Real GDP $Y_1 < Y^*$,
 - Price level and inflation are lower than before.

Step 3: Medium-run adjustment.

- The output gap (Y below potential) eventually puts downward pressure on wages and other input costs.
- SRAS shifts to the right over time, moving the economy back toward potential GDP Y^* at a lower inflation rate.
- The final outcome is output near Y^* and inflation close to the 2% target midpoint.

Summary:

- Contractionary monetary policy: AD shifts left.
- Short run: lower output, higher unemployment, lower inflation.
- Medium run: SRAS shifts right, output returns to potential, inflation stabilizes near target.

Problem 2

We are given the reaction function:

$$i = 1 + 1.5(\pi - 2) + 0.5(Y - Y^*),$$

where i is the target overnight interest rate in percent, π is inflation, and $(Y - Y^*)$ is the output gap in percent.

Data:

$$\pi = 3.4\%, \quad Y - Y^* = +1.5\%.$$

(a) Compute the recommended overnight interest rate

Step 1: Compute the inflation deviation from target.

$$\pi - 2 = 3.4 - 2.0 = 1.4 \quad (\text{percentage points above target}).$$

Step 2: Compute the inflation term $1.5(\pi - 2)$.

$$1.5 \times 1.4 = 2.1.$$

Step 3: Compute the output gap term $0.5(Y - Y^*)$.

$$Y - Y^* = +1.5 \Rightarrow 0.5 \times 1.5 = 0.75.$$

Step 4: Plug values into the reaction function.

$$i = 1 + 1.5(\pi - 2) + 0.5(Y - Y^*) = 1 + 2.1 + 0.75 = 3.85.$$

Answer:

$i = 3.85\% \text{ (target overnight interest rate)}$

(b) Is this expansionary or contractionary? Explain.

Step 1: Compare to a neutral or lower previous rate.

- The rule indicates a relatively *high* policy rate (3.85%) because:
 - inflation is above target (3.4% vs 2%),

- and the output gap is positive (economy above potential).

Step 2: Interpretation.

- Raising the policy rate to 3.85% will *lean against* inflationary pressures and an overheating economy.
- This is a **contractionary** (tightening) monetary policy stance:
 - it is intended to slow down aggregate demand,
 - reduce the positive output gap,
 - and bring inflation back toward 2%.

Conclusion:

The policy stance implied by $i = 3.85\%$ is contractionary.

(c) Effects on lending, long-term rates, AD, and inflation

Step 1: Bank lending and short-term rates.

- A higher overnight rate increases banks' cost of short-term funds.
- Banks typically respond by raising their prime lending rates and other short-term loan rates.
- As borrowing becomes more expensive, households and firms reduce their demand for loans.
- Bank lending slows.

Step 2: Long-term interest rates.

- Expectations of tighter monetary policy raise yields on longer-term bonds.
- Mortgage rates and corporate bond yields may rise.
- This further dampens spending on housing and business investment.

Step 3: Aggregate demand.

- Higher interest rates reduce:
 - **Consumption** of durable goods (cars, appliances, housing-related spending).
 - **Investment** in new capital (machines, buildings, technology).
- The Canadian dollar may appreciate, reducing net exports.
- Combined, C , I , and NX fall, causing a leftward shift in **aggregate demand (AD)**.

Step 4: Inflation in the medium run.

- With AD lower, real GDP growth slows and the positive output gap shrinks or disappears.
- Lower demand pressure reduces upward pressure on wages and prices.
- Over time, this moderates inflation, moving it from 3.4% back toward the 2% target.

Summary:

- Higher policy rate $\Rightarrow$ higher borrowing costs and reduced credit growth.
 - Lower C , I , and $NX \Rightarrow$ lower AD.
 - Lower AD $\Rightarrow$ lower inflation in the medium run.
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End of Solutions